

R Notebook

This is an R Markdown (<http://rmarkdown.rstudio.com>) Notebook. When you execute code within the notebook, the results appear beneath the code.

Try executing this chunk by clicking the *Run* button within the chunk or by placing your cursor inside it and pressing *Ctrl+Shift+Enter*.

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':  
##  
##   filter, lag
```

```
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

#On aggregate level

We use the following equation:

$$PV_{loan} = \sum_{t=1}^T \frac{\alpha \cdot \max\{0, Y_0(1+g)^{t-1} - \tau\}}{(1+r)^t}$$

hence the k is calculated with the following assumptions:

$$\Delta \log(P_i) = \log(22200) - \log \left(\sum_{t=1}^T \frac{\alpha \cdot \max\{0, Y_i(1+g)^{t-1} - \tau\}}{(1+r)^t} \right)$$

where Y_i is the starting salary in country i.

```
# Present value of repayments
# (payments stop when loan is repaid)
pv_loan <- function(Y0,
                    L = 9250,      # loan principal (1 year)
                    alpha = 0.09,  # repayment rate
                    tau = 27295,   # threshold
                    g = 0.02,      # wage growth
                    r = 0.03,      # discount rate
                    T = 30){       # years

  balance <- L
  pv <- 0

  for(t in 1:T){

    # income path
    Yt <- Y0 * (1 + g)^(t - 1)

    # repayment rule
    Rt <- alpha * max(0, Yt - tau)

    # cannot repay more than remaining balance
    payment <- min(Rt, balance)

    # discounted payment
    pv <- pv + payment / (1 + r)^t

    # update remaining loan
    balance <- balance - payment

    # stop once repaid
    if(balance <= 0){
      break
    }
  }

  return(pv)
}
```

```

beta <- -1.28
Ppost <- 22200

elasticity_from_income <- function(Y0){
  PV <- pv_loan(Y0)

  # avoid log(0) for very low incomes
  PV <- max(PV, 1)

  k <- log(Ppost) - log(PV)
  eta <- -beta / k

  return(c(PV = PV, k = k, elasticity = eta))
}

```

```

salary_levels <- c(20000, 26000, 32000, 42000, 32000, 35000, 42000, 50000)

results <- t(sapply(salary_levels, elasticity_from_income))

table_results <- data.frame(
  Starting_Salary = salary_levels,
  PV_Loan = round(results[,1], 2),
  k = round(results[,2], 3),
  Elasticity = round(results[,3], 3)
)

table_results

```

Starting_Salary <dbl>	PV_Loan <dbl>	k <dbl>	Elasticity <dbl>
20000	2366.69	2.239	0.572
26000	5835.48	1.336	0.958
32000	7435.25	1.094	1.170
42000	8301.81	0.984	1.301

Starting_Salary <dbl>	PV_Loan <dbl>	k <dbl>	Elasticity <dbl>
32000	7435.25	1.094	1.170
35000	7822.44	1.043	1.227
42000	8301.81	0.984	1.301
50000	8547.92	0.954	1.341

8 rows

```
library(ggplot2)

beta <- -1.28
Ppost <- 22200

PV <- seq(500, 9250, by = 50)
elasticity <- -beta / (log(Ppost) - log(PV))
curve_df <- data.frame(PV, elasticity)

# ---- Loan valuation function ----
pv_loan <- function(Y0,
                    L = 9250,
                    alpha = 0.09,
                    tau = 27295,
                    g = 0.02,
                    r = 0.03,
                    T = 30){

  balance <- L
  pv <- 0

  for(t in 1:T){
    Yt <- Y0 * (1 + g)^(t - 1)
    Rt <- alpha * max(0, Yt - tau)
    payment <- min(Rt, balance)
    pv <- pv + payment/(1+r)^t
    balance <- balance - payment
    if(balance <= 0) break
  }

  return(max(pv,1))
}

# only representative income groups (not 10 lines)
salary_levels <- c(20000, 26000, 32000, 42000)

salary_df <- data.frame(
  salary = salary_levels,
  PV = sapply(salary_levels, pv_loan)
```

```
)

salary_df$elasticity <- -beta/(log(Ppost)-log(salary_df$PV))

# ---- plot ----
ggplot(curve_df, aes(PV, elasticity)) +

  # main curve
  geom_line(linewidth=1.4, color="#E66100") +

  # benchmark tuition
  geom_vline(xintercept = 9250,
             linetype="dashed",
             color="red",
             linewidth=1) +

  # salary points
  geom_point(data=salary_df,
            aes(PV, elasticity),
            size=3.5,
            color="black") +

  # labels above points
  geom_text(data=salary_df,
           aes(PV, elasticity, label=paste0("£",salary)),
           nudge_y=0.08,
           size=4) +

  labs(
    x="Economic value of the UK student loan (Present value, £)",
    y="Implied demand elasticity",
    title="How valuable was the UK student loan to EU applicants?",
    subtitle="Points show elasticity implied by expected graduate earnings"
  ) +

  coord_cartesian(ylim=c(0.3,1.6)) +

  theme_classic(base_size=15) +
  theme(
```

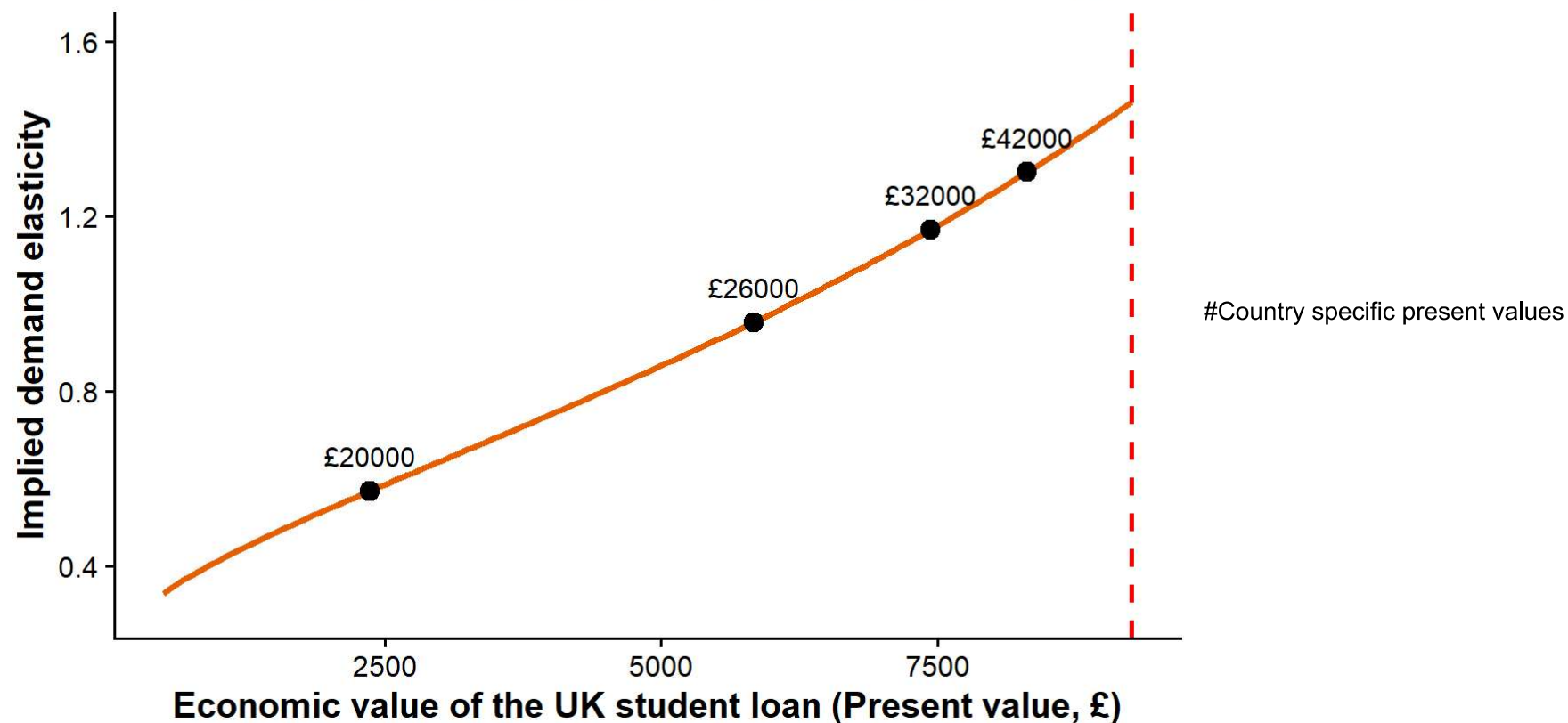
```

plot.title = element_text(face="bold", hjust=0.5),
plot.subtitle = element_text(hjust=0.5),
axis.title = element_text(face="bold")
)

```

How valuable was the UK student loan to EU applicants

Points show elasticity implied by expected graduate earnings



```

library(readxl)
wages<-read_excel("C:/Users/Stoycho.rusinov/Desktop/Brexit_paper/Present_value_simulations/wages.xlsx", sheet="main_df")

```